FBG touch sensors in a social robot: Toward natural behaviors controlled by physical Human-Robot Interactions

Elizabeth Sanchez R[a], María Gaitán-Padilla [a], Vinicius Mageste Coimbra [a], Maria José Pontes [a], Marcelo Eduardo Vieira Segatto [a], Marcela Munera[b], Carlos A. Cifuentes[b], and Camilo A. R. Diaz*[a]

[a] Telecommunications Laboratory (LabTel), Electrical Engineering Department, Federal University of Espírito Santo, Vitoria, Brazil;

[b] Bristol Robotics Laboratory, University of the West of England, Bristol, UK.

ABSTRACT

Social assistive robots (SARs) have been widely applied to improve users' quality of life by encouraging continuous and natural human-robot interactions (HRIs). The robot's ability to effectively perceive and respond to user actions is essential in achieving these interactions. Sensors play a pivotal role in ensuring these behaviors align with the dynamic needs of users, particularly in sensitive contexts like therapeutic settings. This work presents the development and initial validation of a Fiber Bragg Grating (FBG)-based sensor for touch detection in the hand of a SAR, aimed at maintaining HRIs during therapy. The proposed system involves the integration of FBG sensors into four segments of the robot's hand, enabling precise detection of the touch area. A convolutional neural network (CNN) was employed to classify the touch data, demonstrating accurate performance in identifying contact areas across the hand's surface. The sensor's robustness and adaptability to the robot's instrumentation were also evaluated, highlighting its potential to operate reliably in therapeutic environments. This generates future research into interaction modalities, such as multi-point contact detection and adaptive interaction responses sensed with optical fiber-based technology.

Keywords: Fiber Bragg Gratings, Human-Robot Interaction, Optical fiber sensors, Touch Detection.

1. INTRODUCTION

In recent decades, Socially Assistive Robots (SARs) have emerged as a promising technology to enhance the quality of life for individuals in various contexts, from healthcare to home assistance [1]. Human-Robot Interaction (HRI) plays a crucial role in the success of these applications, enabling more natural and effective communication between users and robotic systems [2]. In particular, physical human-robot interaction (pHRI), which involves physical contact, has gained significant attention due to its crucial role in social interaction [3].

Implementing advanced sensors in pHRI systems is essential to ensure the safety and effectiveness of interactions. Various types of sensors, such as tactile, force/torque, and proximity sensors, have been used in robotic platforms to detect and respond to physical interactions [4]. As SARs evolve to perform more complex tasks and interact in dynamic environments, there is a growing demand for sensing technologies that offer precision, reliability, and adaptability, as well as the deployment of machine learning algorithms. In assistance cases, the robot understands the human intention and reacts to carry out the planned task, being more natural due to environmental sensing [5].

The optical-fiber-based sensors have been used in different sensing applications, by using light, for instance, wavelength [6], or intensity variation systems [7], and being small and potentially used in wearable and embedded devices [6], representing a recent area for pHRI detection. Some optical fiber sensors have been used in collaborative robots, detecting objects, materials, and movements. Massari *et al.* developed a tactile skin for a robotic arm with optical fiber sensing, especially using Fiber Bragg Gratings (FBG) [8]. This technology works with a grating of refraction indexes in the fiber's core, generating a returning light signal. The variations in the segment of the FBG generate a shift in the returned wavelength, which can be monitored with an interrogation system [9].

Concerning HRIs, the objective is to develop new HRI strategies with a SAR developed for therapies through FBG sensors. In the area of SAR, the CASTOR project started with the inclusion of robotics in therapeutic environments [10], as well as, the lack of implementations in Latin American countries [11]. The aim is to develop an FBG-based sensor system that allows the detection of pHRIs in the robot's hand to control the robot's actions according to the HRIs maintained and their location.

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2. METHODOLOGY

This section presents the FBG sensor developed and adapted to the hand of the SAR, the setup proposed to identify the touching areas, the procedure, data acquisition, and the data analysis and management for the convolutional neural network (CNN) made for the classification.

The FBG sensor

The FBG sensor was made using an ultraviolet (UV) laser and phase mask technique. This process involves the inscription of periodic refractive index changes along the core of a single-mode optical fiber to create the grating structure [9]. The procedure began by preparing two segments of boron-doped photosensitive optical fiber (*FiberCore, United Kingdom*), removing the coating along the section designated for FBG inscription, and the regions were cleaned with isopropyl alcohol for efficient UV exposure. Then, two phase masks (*Ibsen Photonics, Denmark*) were used to inscribe the desired Bragg-centered wavelengths by positioning them parallel to the fiber. The UV laser (*CNI DPS-266, China*) with an operating wavelength of 266 nm and nanosecond pulses of 8 ns with a repetition rate of 100 Hz created an interference pattern, inducing periodic refractive index modulation in the fiber core [6]. Alignment ensured uniform exposure, while laser fluence and exposure time were controlled with the optical sensing interrogator SM125 (*Micron Optics, Inc.*) to achieve the desired grating peaks. After the inscription, the fibers were coated with DeSolite 950-200 (*Fiber Optic Center, Inc, United States*) for protection with the Vytran Fiber Recoater (*Thorlabs, United States*). Finally, a refractive index gel was applied to the fiber's end to reduce undesirable light reflection. The grating's spectral characteristics were verified using the optical interrogator system.

The inscribed FBGs were embedded into a 3D-printed guide. The sensitive areas of the FBGs were exposed to be covered with a layer of EcoFlex silicone (*Polisil, Brazil*) for protection and to ensure the sensing area. The fiber was secured using plasticine to be centered in the guide ensuring variations when applying strain and touch. Each segment of the robot's hand was instrumented with a dedicated FBG sensor, in total 4 FBGs in two optical fiber guides were made, distributed in the robot as shown in Figure 1b (FBG1 at the front and high (T1), FBG2 at the front down (T2), FBG3 at the back high (T3), and FBG4 at the back down (T4)). The centered Bragg wavelengths of the four FBGs are shown in Figure 1c, being T1 and T3 at 1548.3 nm and T2 and T4 at 1562.6 nm.

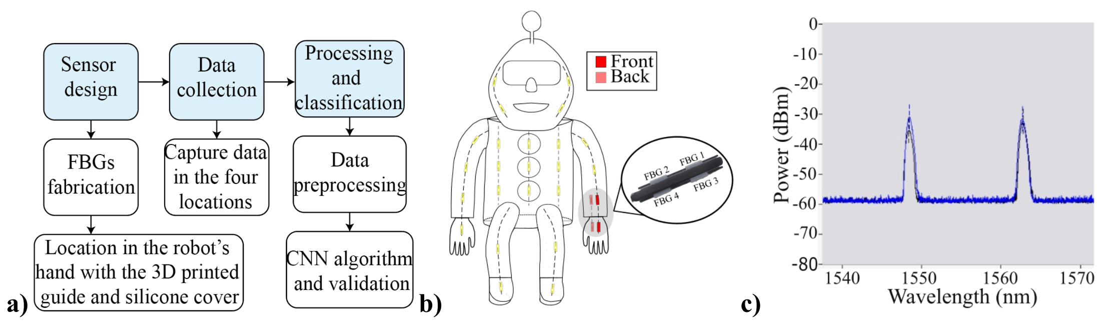

Figure 1. Methodology overview and sensor configuration. a) Methodology diagram, b) The setup proposed with FBG sensors in the robot highlighting in red the four FBGs in the hand, and c) Bragg wavelengths of the four developed FBGs.

Procedure

For capturing data, 10 segments of five seconds for each hand area were recorded at 1000 Hz with the Hyperion Si255 interrogator (*Micron Optics, Inc.*) and the ENLIGHT Software (*Micron Optics, Inc.*). Each data segment had a different quantity and intensity of touches, adding naturality to the interactions and complexity to the algorithm.

Data analysis and the algorithm

The acquired signals were preprocessed using a moving average filter to remove periodic trends. This was implemented in MATLAB R2024b (*MathWorks, USA*) using the smoothdata function, which applies a smoothing factor to determine the window size. Various values were tested, ranging from 0.1 to 0.6. A value of 0.25 was selected providing the best balance between noise reduction and data fidelity. Then, a CNN was used for the data classification of the touching area, detailed in the Results section. The CNN architecture consisted of two Conv1D layers with a ReLU activation function for feature extraction, dropout layers for regularization, and a dense output layer with softmax activation for classification. The dataset was divided into training and validation sets with an 80:20 ratio, and it used a 5-fold cross-validation strategy to avoid overfitting. The model was compiled using the Adam optimizer and sparse categorical cross-entropy loss, balancing performance, and generalization.

3. RESULTS

This section presents the FBG sensor developed and integrated into the hand of the CASTOR robot, the results of the signal characterization, and the identification of the touching areas with a CNN.

The FBG sensor

Figure 2a shows the final setup of the FBG sensor integrated into the hand of the CASTOR robot, demonstrating its adaptability to the robotic system. Figure 2b shows an example of the signals obtained when the sensor was subjected to touch interactions in T3.

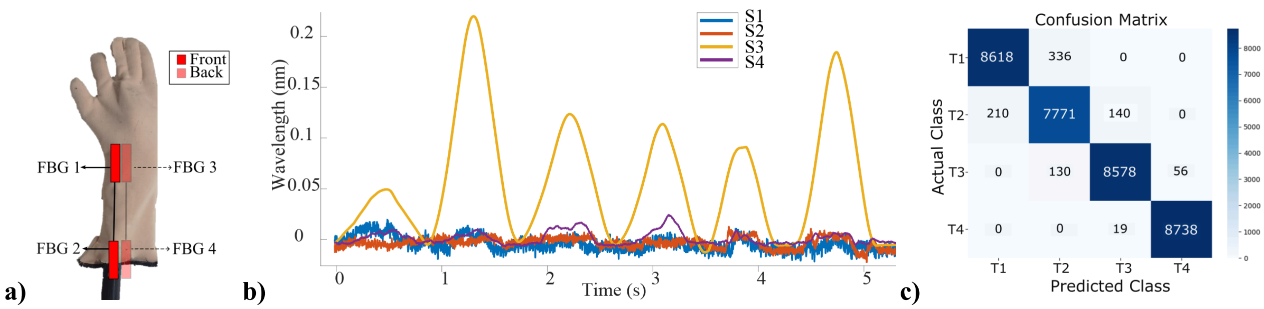

Figure 2. Overview of sensor deployment and evaluation. a) FBG sensor deployed in the robot's hand, b) example of the FBGs responses by touching section 3, and c) confusion matrix of the CNN model.

Table 1. CNN Model Evaluation Metrics.

Metric	T1	T2	T3	T4
Accuracy	0.93	0.97	0.99	1
Recall	1	0.91	0.98	1
F1-Score	0.96	0.94	0.98	1

Sensor's classification

Figure 2c presents the confusion matrix of the CNN model, indicating high classification accuracy across all classes. The diagonal dominance of the matrix reflects the model's ability to correctly predict the major part of instances and robust performance in distinguishing between different touching areas. The evaluation metrics for the CNN model are summarized in Table 1. Results demonstrate the sensor's accuracy in classifying the four touching areas. Each area (T1–T4) achieved accuracies ranging from 0.93 to 1.00 and F1-scores above 0.94. The recall values indicate that the sensor and model captured true positive events, with a recall of 1.00 observed for touch zones T1 and T4. It reflects the model's balanced performance across all classes.

4. DISCUSSION

The proposed implementation of four multiplexed FBG sensors for robotic touch classification was due to strengths in employing FBG sensors [6], which are lightweight, easily embedded in devices, and immune to electromagnetic interference. In the proposed setup the system achieves accurate and reliable touch detection on the robot's hand. The integration of two sensors per optical fiber optimizes spatial distribution while maintaining a compact design, which is critical for robotic applications requiring touch sensitivity [8].

Additionally, using a CNN for touch classification achieves high-performance validation across all evaluated metrics. These results indicate that the CNN effectively learns and distinguishes the signal patterns corresponding to each touch area, as previously evaluated by Lyu *et al.* with FBG sensors and a Gramian angular field algorithm with a CNN, obtaining accuracies of 99.75 in six object types [9]. The recall value of 0.91 in T2 highlights a slightly higher rate of missed classifications for that area, which could result from variations in strain due to differences in the silicone cover.

Nevertheless, the overall performance remains robust, particularly given the simple preprocessing applied to the signals, this approach leads to future work with more touching points and more complex HRI patterns.

A viscoelastic silicone cover was proposed to enhance the robustness and performance of the FBG sensors, as used in more FBG applications [6, 8], providing mechanical protection and improved strain distribution. The silicone layer ensures that the deformation is spread over a larger area and not directly in the fiber, which enhances sensor durability and reduces the risk of mechanical damage. Additionally, the viscoelastic properties of silicone contribute to better signal stability by smoothing the pHRIs. However, while the cover improves robustness, it may slightly reduce the sensitivity to fine touches or low variations, as the silicone acts as a buffer between the contact point and the FBG.

However, the implementation also presents certain limitations. Firstly, the mechanical integration of FBG sensors in the robotic hand is subject to alignment challenges and strain transfer dependent on the silicone deformation and influence in the FBG, which was different for the four areas and could affect repeatability. Environmental factors such as temperature fluctuations may also influence FBG readings [6], which can be a problem in robotic systems that use electronics that heat in time. Finally, while the current configuration achieves high accuracy, the interrogation system is large, resulting in space and weight occupation in the robotic system. Smaller commercial interrogators are available to achieve a more portable sensing system, moreover, increasing the number of touch points may require more multiplexed FBGs or more reading channels, for which these portable interrogators are limited in wavelength ranges, channels, and cost.

5. CONCLUSION AND FUTURE WORK

This study developed an FBG-based sensor system adapted to a robot's hand, the system was capable of detecting pHRIs and classifying their locations using a CNN. This was achieved by integrating four multiplexed FBG sensors distributed across two optical fibers and enhancing robustness with a viscoelastic silicone cover, the system achieved high classification accuracy, recall, and F1-score values. These results validate the system's ability to accurately map touch areas, demonstrating its potential for reliable and robust pHRI detection. Using a CNN trained with preprocessed signals simplifies the system's input data while performing the classification. Future work will focus on addressing current limitations and expanding system capabilities. First, the alignment and strain transfer inconsistencies across different zones will be further investigated to improve repeatability and uniform sensor performance. Additionally, it is intended to increase the number of touchpoints by integrating more multiplexed FBGs or employing alternative interrogation systems. Finally, testing the system in more complex pHRI scenarios, such as dynamic force detection or interaction patterns, will enable adaptive robotic control and validate the system’s applicability in real-world assistive robotic tasks.

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